

GLOBAL HEALTH ANALYTICS

A Power BI Business Intelligence Report

Disease Burden, Healthcare Access & Outcomes, 2000–2024

Group 9

Bophine Arnold Odiyo 668821

Lavender Achieng' Onyango 671809

Susan Otieno 670501

Angie Mwanzia 663887

Reana Auma 669255

Movine Rading 670351

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1. Executive Summary

This report documents the design, development, and findings of a Power BI business intelligence solution built on a global health dataset covering 20 countries, 20 diseases, and 25 years of records (2000–2024). The solution transforms 1,000,000 raw rows into a governed star-schema data model, 18 DAX measures, and a 5-page interactive dashboard supporting executive reporting, trend analysis, geographic comparison, and detailed drill-through investigation.

The dashboard is intended to support a global health analytics function in identifying disease burden, monitoring mortality and recovery outcomes, and assessing healthcare system capacity across countries in order to prioritize intervention and funding decisions.

2. Business Problem and Data Understanding

2.1 Business Problem

Global health systems face rising disease burden, uneven healthcare access, and constrained resources. Decision-makers within health ministries and international health bodies need a consolidated, data-driven view of disease trends, mortality and recovery outcomes, and healthcare capacity across countries in order to prioritize funding and interventions effectively.

2.2 Target Organization

A global health analytics function - modelled on a WHO-style monitoring unit or a national Ministry of Health - that benchmarks its own outcomes against 19 peer countries to identify where investment and policy attention are most needed.

2.3 Dataset Source

Global_health_data.csv - a global health records dataset used for this project, containing 1,000,000 rows and 21 source columns covering 20 countries and 20 diseases from 2000 to 2024. (Insert the exact source/download link here, e.g. the Kaggle dataset page, to complete this section for submission.)

2.4 Dataset Description

The source file arrived as a single flat table with one row per country–year–disease–demographic combination, containing:

- Identifiers: Country, Year, Disease Name, Disease Category, Age Group, Gender, Treatment Type
- Outcome measures: Prevalence Rate (%), Incidence Rate (%), Mortality Rate (%), Recovery Rate (%), DALYs, Improvement in 5 Years (%)
- Volume measures: Population Affected
- Healthcare capacity: Healthcare Access (%), Doctors per 1000, Hospital Beds per 1000, Availability of Vaccines/Treatment, Average Treatment Cost (USD)
- Socio-economic context: Per Capita Income (USD), Education Index, Urbanization Rate (%)

This flat table was restructured in Power Query and Power BI into a fact table plus six supporting tables

2.5 Key Business Questions

- Which diseases and disease categories carry the highest mortality relative to recovery, and where?
- Which countries are underperforming on healthcare access and patient outcomes?
- What trends in prevalence, incidence, and treatment cost are visible across 2000–2024?
- What risks - low doctors-per-1000, low vaccine availability, high mortality — should management address first?
- Where do income, education, and urbanization correlate with better health outcomes, indicating opportunities for targeted investment?

3. Data Preparation (Power Query)

The raw 1,000,000-row extract was cleaned and reshaped in Power Query before loading into the data model. Core cleaning steps applied:

- Renamed unclear columns for consistency (e.g. shortened long source headers to model-friendly names such as MortalityRate, RecoveryRate, HealthcareAccess, VaccineAvailable).
- Corrected data types - Year as whole number; all rate and percentage fields as decimal; Population Affected and DALYs as whole numbers; categorical fields as text.
- Removed duplicate and blank rows.
- Trimmed and cleaned text columns (Country, Disease Name) to remove whitespace and casing inconsistencies.

- Replaced inconsistent categorical values and standardized Yes/No fields.
- Handled missing values using column profiling to identify nulls before substitution.
- Removed columns not required for the model scope.
- Created conditional/custom columns, including a Disease Category lookup, a recovery-rate flag, a treatment cost category, and a high-risk flag (RecoveryRateFlag, CostCategory, IsHighRisk in the delivered model).
- Extracted year (and derived date) parts to support time intelligence.

3.1 Advanced Power Query Techniques Applied

- Reference queries used to derive Dim_Country, Dim_Disease, Dim_Demographics, and Dim_Treatment from the cleaned source query without duplicating the load.
- Group By with multiple aggregations used to build the Summary_by_Country_and_Disease table (pre-aggregated by country and disease).
- A dedicated Date table created and connected as the model's time dimension.
- Business categories created using nested conditional logic (e.g. CostCategory, IsHighRisk).
- Column profiling used throughout (profiled on the full 1,000,000-row dataset) to check distribution and quality before finalizing transformations.
- Error handling applied to numeric columns following type conversion (a TestErrorCheck field remains in the model as evidence of this validation step).

Evidence to attach for submission: screenshots of the raw data, the Power Query Editor, the Applied Steps pane, Column Profiling view, the final cleaned table, and one advanced transformation (e.g. the Group By step building Summary_by_Country_and_Disease).

4. Data Modelling

The model follows a star-schema design with one central fact table and six supporting tables, avoiding a single flat table.

4.1 Tables in the Model

Table	Role	Description
Fact_HealthRecords	Fact table	1,000,000 rows; core measures (Population Affected, MortalityRate, RecoveryRate, AvgTreatmentCost, DALYs, etc.) plus key columns for relating to dimensions.
Dim_Country	Dimension	Unique list of 20 countries.
Dim_Disease	Dimension	Unique Disease Name / Disease Category combinations (20 diseases).
Dim_Date	Dimension (Date table)	Calendar table spanning 2000–2024, marked as the model's official Date table.
Dim_Demographics	Dimension	Unique Age Min/Max and Gender combinations.
Dim_Treatment	Dimension	Unique Treatment Type values.
DiseaseCategoryLookup	Supporting lookup	Maps Disease Name to Disease Category for consistent categorization.
Summary_by_Country_and_Disease	Aggregation table	Pre-aggregated Group By output by Country and Disease, used to speed up summary visuals.

4.2 Relationships

Key relationships connecting the fact table to its dimensions:

- Dim_Country[Country] → Fact_HealthRecords[Country] (one-to-many)
- Dim_Disease[Disease Name] → Fact_HealthRecords[Disease Name] (many-to-many, both-direction filtering to support the DiseaseCategoryLookup pattern)
- Dim_Treatment[Treatment Type] → Fact_HealthRecords[Treatment Type]

- Dim_Demographics[AgeMax] → Fact_HealthRecords[AgeMax]
- Dim_Date[Date] → Fact_HealthRecords[Date] (one-to-many, single direction — powers all time intelligence measures)
- Fact_HealthRecords[Country] and Disease Name also link into Summary_by_Country_and_Disease and DiseaseCategoryLookup respectively, supporting the pre-aggregated and lookup tables.

Dim_Date is marked as the official Date table, satisfying the star-schema time-intelligence requirement.

4.3 Star Schema Confirmation

The fact table sits at the centre with all six supporting tables surrounding it — Country, Disease, Date, Demographics, Treatment, and the Disease Category lookup — meeting the requirement to avoid one large flat table.

5. DAX Measures

18 DAX measures were created, covering all required categories: basic aggregation, time intelligence, percentage/ratio, ranking, conditional KPI/status, and dynamic titles.

5.1 Basic Aggregation

Measure	DAX Expression
Total Cases	SUM(Fact_HealthRecords[Population Affected])
Avg Mortality Rate	AVERAGE(Fact_HealthRecords[MortalityRate])
Avg Recovery Rate	AVERAGE(Fact_HealthRecords[RecoveryRate])
Total Treatment Cost	SUM(Fact_HealthRecords[AvgTreatmentCost USD])
Distinct Diseases	DISTINCTCOUNT(Fact_HealthRecords[Disease Name])

5.2 Time Intelligence

Measure	DAX Expression
Cases PY	CALCULATE([Total Cases], SAMEPERIODLASTYEAR(Dim_Date[Date]))

Measure	DAX Expression
Cases YoY %	DIVIDE([Total Cases] - [Cases PY], [Cases PY])
Mortality Rate YTD	TOTALYTD([Avg Mortality Rate], Dim_Date[Date])
Cases 3Y Avg	AVERAGEX(DATESINPERIOD(Dim_Date[Date], MAX(Dim_Date[Date]), -3, YEAR), [Total Cases])

5.3 Percentage and Ratio

Measure	DAX Expression
Recovery vs Mortality Ratio	DIVIDE([Avg Recovery Rate], [Avg Mortality Rate])
% of Global Cases	DIVIDE([Total Cases], CALCULATE([Total Cases], ALL(Dim_Country)))
Vaccine Coverage %	DIVIDE(CALCULATE([Total Cases], Fact_HealthRecords[VaccineAvailable]="Yes"), [Total Cases])

5.4 Ranking

Measure	DAX Expression
Country Rank by Mortality	RANKX(ALL(Dim_Country[Country]), [Avg Mortality Rate], , DESC)
Disease Rank by Cases	RANKX(ALL(Dim_Disease[Disease Name]), [Total Cases], , DESC)

5.5 Conditional KPI / Status

Measure	DAX Expression
Mortality KPI Status	IF([Avg Mortality Rate] > 10, "Critical", IF([Avg Mortality Rate] > 5, "Watch", "Stable"))
Healthcare Access KPI	IF(AVERAGE(Fact_HealthRecords[HealthcareAccess]) >= 80, "Good", "Needs Attention")

5.6 Dynamic Titles

Measure	DAX Expression
Dynamic Title Country	"Health Overview: " & SELECTEDVALUE(Dim_Country[Country], "All Countries")
Dynamic Title Disease	"Analysis: " & SELECTEDVALUE(Dim_Disease[Disease Name], "All Diseases")

6. Dashboard Design and Visuals

The dashboard is delivered across five pages, using a light corporate theme suited to a health-sector audience.

6.1 Executive Summary

Header: “Global Health Statistics Dashboard”. Contains four KPI cards (Total Cases, Total Treatment Cost, Avg Mortality Rate, Avg Recovery Rate), a line chart of trends over time, a clustered bar chart, and three synced slicers for filtering by country, disease, and year.

6.2 Trend Analysis

Header: “Disease Trend Analysis”. Contains a stacked area chart and line chart tracking outcomes across 2000–2024, a ribbon chart showing category rank shifts by year, a pivot table for detailed year-by-year figures, and a slicer for interactive filtering.

6.3 Geographic Analysis

Header: “Geographic Disease Analysis”. Contains a filled/Azure map for country-level comparison, a treemap for category and disease-level breakdown, a scatter chart plotting relationships between socio-economic and outcome measures, and a slicer for interactivity.

6.4 Detailed Analysis

Header: “Patient & Disease Detailed Analysis”. Contains a detailed table (tableEx), a decomposition tree for interactive breakdown of cases by country, disease, and demographic, an action button for navigation, and a slicer.

6.5 Tooltip Page

A hidden report-page tooltip containing a compact line chart, displayed on hover over key visuals on the main pages to give viewers quick contextual trend detail without navigating away.

6.6 Design Elements Applied

- Consistent colour palette and professional theme across all pages
- Dynamic, filter-responsive page titles bound to Dynamic Title Country / Dynamic Title Disease
- Conditional formatting applied to KPI and status fields (Mortality KPI Status, Healthcare Access KPI)
- Synced slicers for Country, Disease, and Year across report pages
- Drill-through and custom tooltip functionality for deeper investigation without cluttering the main pages

7. Key Insights from the Data

Aggregating all 1,000,000 records produces the following headline figures:

Metric	Value
Total population-affected instances recorded	500.7 billion (sum across all rows)
Average Mortality Rate	5.05%
Average Recovery Rate	74.50%
Total Treatment Cost recorded	\$25.0 billion (sum across all rows)
Countries covered	20
Diseases covered	20
Years covered	2000–2024 (25 years)
Vaccine/treatment availability	50.0% of records marked "Yes"

Country and disease comparisons: Turkey, France, and the UK show the highest average mortality rates in the dataset (all approximately 5.06–5.07%), while Zika, Cancer, and Leprosy account for the highest total case volumes by disease. Metabolic, Neurological, and Parasitic categories carry the largest share of total cases by disease category.

An important, honest observation for the report: across country, disease, gender, treatment type, and year, the differences in mortality rate, recovery rate, and case volume are consistently small (typically well under one percentage point), and correlations between socio-economic fields (income, healthcare access, urbanization) and outcomes (mortality, recovery) are close to zero. This indicates the underlying dataset is evenly/randomly distributed rather than reflecting real-world epidemiological skew. For the presentation, frame country/disease "leaders" as the relative ranking produced by the model (which is exactly what the rubric's ranking measures are designed to demonstrate) rather than claiming large real-world disparities - this is an accurate and defensible way to present a synthetic practice dataset.

8. Recommendations

- Prioritize monitoring of the disease categories and countries flagged "Critical" or "Watch" by the Mortality KPI Status measure for closer policy attention.
- Use the Healthcare Access KPI to identify countries below the 80% access threshold as candidates for infrastructure investment (additional doctors and hospital beds per 1000).
- Track Vaccine Coverage % over time to identify where treatment/vaccine availability gaps persist and prioritize distribution accordingly.
- Use the Country Rank by Mortality and Disease Rank by Cases measures on the Executive Summary page as a standing watchlist that updates automatically as new data is loaded.

9. Conclusion

This project delivered a complete, governed Power BI solution - from a single raw CSV extract to a star-schema model, 18 DAX measures across all required categories, and a 5-page interactive dashboard with drill-through and tooltip functionality. The solution directly answers the business questions set out in Section 2.5 and provides a reusable framework for ongoing global health monitoring.